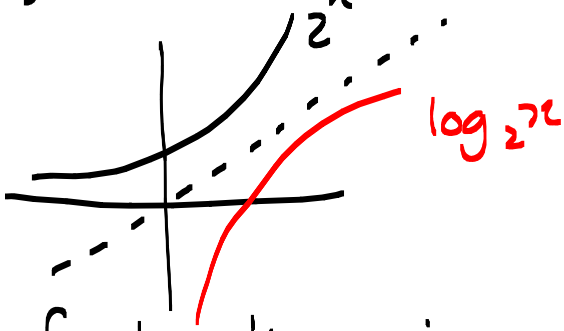
Unit 1.9 Logarithmic functions. p 59

Note Title 2015/02/10

$f(x) = 2^x$ is a one-to-one function



To find the inverse of $f(x) = 2^x$

1. $y = 2^x$

$D_f = \mathbb{R}$
$R_f = (0, \infty)$

2. $x = 2^y$

3. $y = \log_2 x$ $\therefore$ $f^{-1}(x) = \log_2 x$

$D_{f^{-1}} = (0, \infty)$
$R_{f^{-1}} = \mathbb{R}$

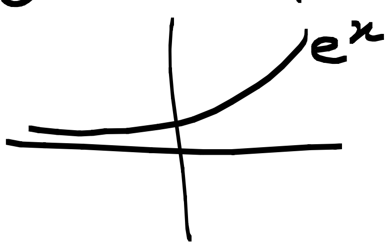
Definition:

$$\boxed{\log_a x = y \iff x = a^y}$$

Laws of logs. p 60.

Natural log.

$f(x) = e^x$ 1-to-1 function

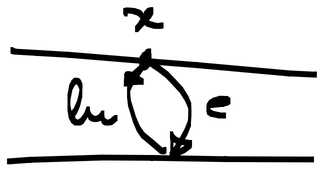


$$f^{-1}(x) = \log_e x = \underbrace{\ln}_{} x$$

$$\boxed{\ln x = y \iff e^y = x.}$$

<u>Logarithmic rules:</u>

1. $\ln(ab) = \ln a + \ln b$
2. $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$
3. $\ln(a^r) = r \ln a$



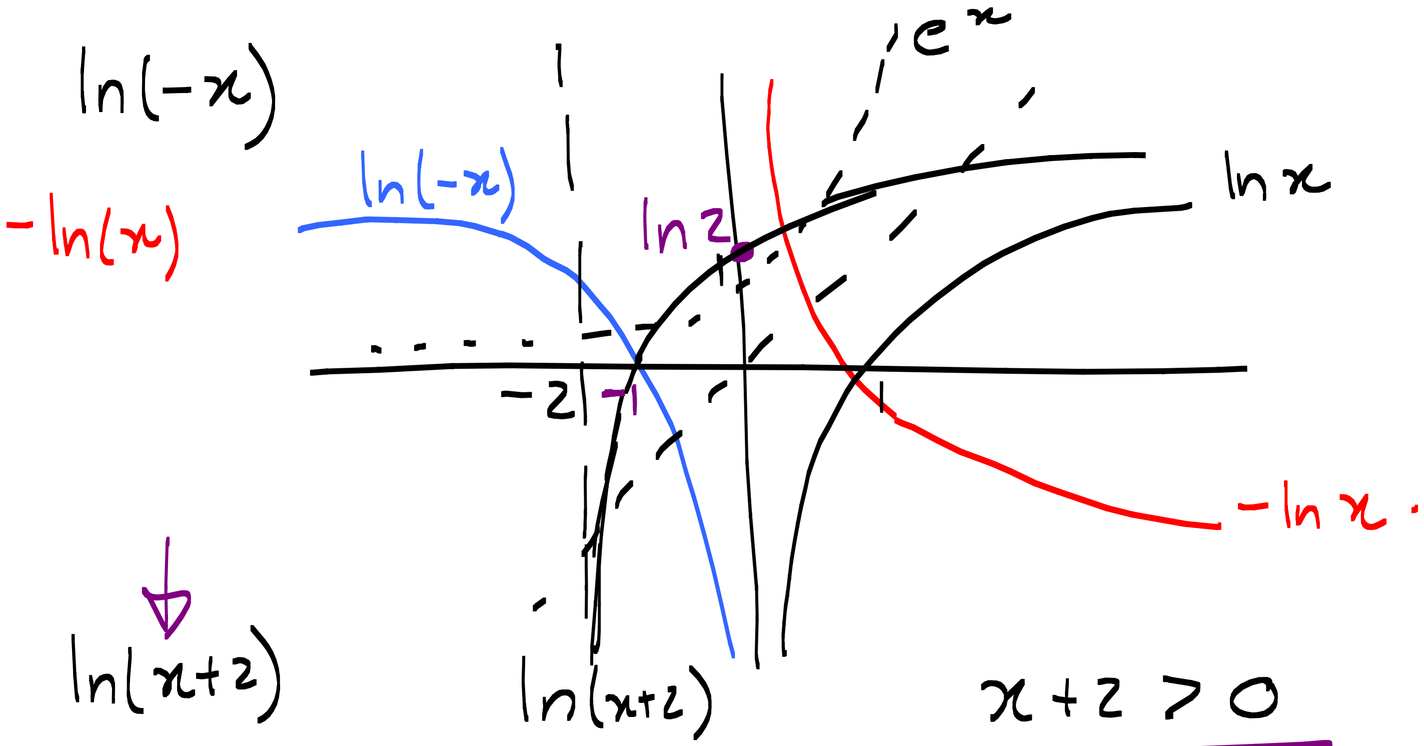
$$a, b > 0$$

4. $\ln e^x = x$
5. $\ln e = 1$
6. $e^{\ln x} = x$

$$\log_e e = 1$$

$$e^{2\ln x} = e^{\ln x^2} = x^2 \ !!$$

Sketch $y = \ln x$



$x + 2 > 0$

$$\boxed{x > -2}$$

$y = \ln(x+2)$ $\qquad$ $D = (-2, \infty)$

$x = 0 \Rightarrow y = \ln 2$

$y = 0 \Rightarrow 0 = \ln(x+2)$

$e^0 = e^{\ln(x+2)}$ $\qquad$ e ln 1-to-1

$1 = x + 2$

$\Rightarrow x = -1$

① Simplify $\ln\left(\frac{1}{e}\right) = \ln 1 - \ln e$

$= 0 - 1 = -1$

② Solve $e^x = 9e^{3x}$

$\ln e^x = \ln(9e^{3x})$

$x \ln e = \ln 9 + \ln e^{3x}$

$$x = \ln 9 + 3x \ln e$$
$$-2x = \ln 9$$
$$x = \frac{2 \ln 3}{-2} = -\ln 3$$
$$= \ln 3^{-1}$$
$$= \ln\left(\frac{1}{3}\right)$$

3. Solve for x:

$$e^{5-3x} = 10$$
$$\ln\left(e^{5-3x}\right) = \ln 10$$
$$5 - 3x = \ln 10$$
$$x = \frac{\ln 10 - 5}{-3}.$$

4. $\ln 2x = 4 + 3 \ln x \quad , \quad x > 0$

$$\ln 2x - 3\ln x^3 = 4$$
$$e^{\ln \frac{2x}{x^3}} = e^4$$
$$\frac{2x}{x^3} = e^4$$
$$\frac{2}{x^2} = e^4 \Rightarrow x^2 = \frac{2}{e^4}$$
$$\Rightarrow x = +\sqrt{\frac{2}{e^4}}$$

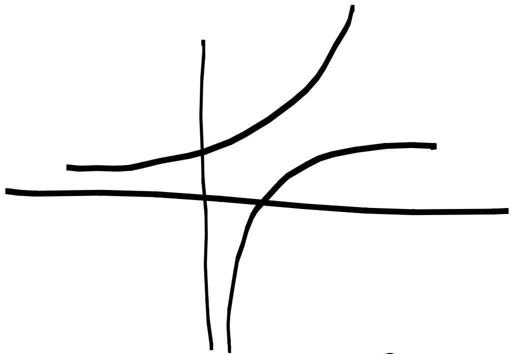
5. $\ln\left[\dfrac{x^3 e^{x^2}}{\sqrt{x}}\right]$ Simplify.

$$= \ln x^3 + \ln e^{x^2} - \ln\sqrt{x}$$

$$= 3\ln x + x^2 - \tfrac{1}{2}\ln x.$$

$$= \tfrac{5}{2}\ln x + x^2$$

Inequalities



def $\begin{cases} \text{1-to-1 function} \\ \text{increasing} \end{cases}$

1. Solve $\ln x < 3$

$$e^{\ln x} < e^3.$$

$$x < e^3$$

and $x > 0$

$$x \in (0, e^3)$$

$x \in \mathbb{R}$

2. $$e^{2-3x} > 4$$

$$\ln\left[e^{2-3x}\right] > \ln 4$$

$$(2-3x)\ln e > \ln 4$$

$$-3x > \ln 4 - 2$$

$$x < \frac{2-\ln 4}{3}$$

$$x \in \left(-\infty, \frac{2-\ln 4}{3}\right)$$

3. Find the inverse of $f(x) = 1 + e^{-2x}$ and sketch $f^{-1}(x)$ and $f(x)$.

$\ln AB = \ln A + \ln B$.

1. $y = 1 + e^{-2x}$

2. $x = 1 + e^{-2y}$

$\underline{\ln x = \ln(1 + e^{-2y})}$

No good!

3. $x - 1 = e^{-2y}$

$\Rightarrow \ln(x-1) = \ln e^{-2y}$

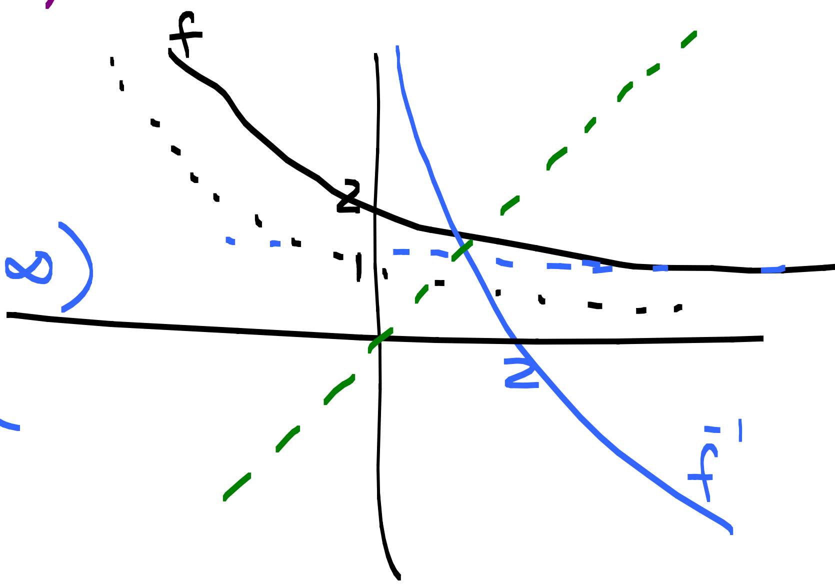
$\Rightarrow \ln(x-1) = -2y \Rightarrow$

$y = -\frac{1}{2}\ln(x-1) \Rightarrow f^{-1}(x) = -\frac{1}{2}\ln(x-1)$

$y = 1 + e^{-2x}$

$D_f = \mathbb{R}$ $\quad R_f = (1, \infty)$

$D_{f^{-1}} = (1, \infty)$ $\quad R_{f^{-1}} = \mathbb{R}$



__Notation:__

$(f \circ f^{-1})(x) = f(f^{-1}(x)) = x$ if $x \in D_{f^{-1}}$

$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = x$ if $x \in D_f$

$f(x) = e^x \qquad f^{-1}(x) = \ln x$

$(f \circ f^{-1})(x) = x$ if $x > 0$

$(f^{-1} \circ f)(x) = x$ if $x \in \mathbb{R}$

Domain: $\frac{1}{\bigcirc}$, $\sqrt{\bigcirc}$, $\ln \bigcirc$

ST 1

Extra example:

Solve $\quad 2 < \ln x < 9$

$\Rightarrow \quad e^2 < x < e^9$

$\Rightarrow \quad x \in (e^2, e^9)$

> e is an increasing 1-to-1 function!